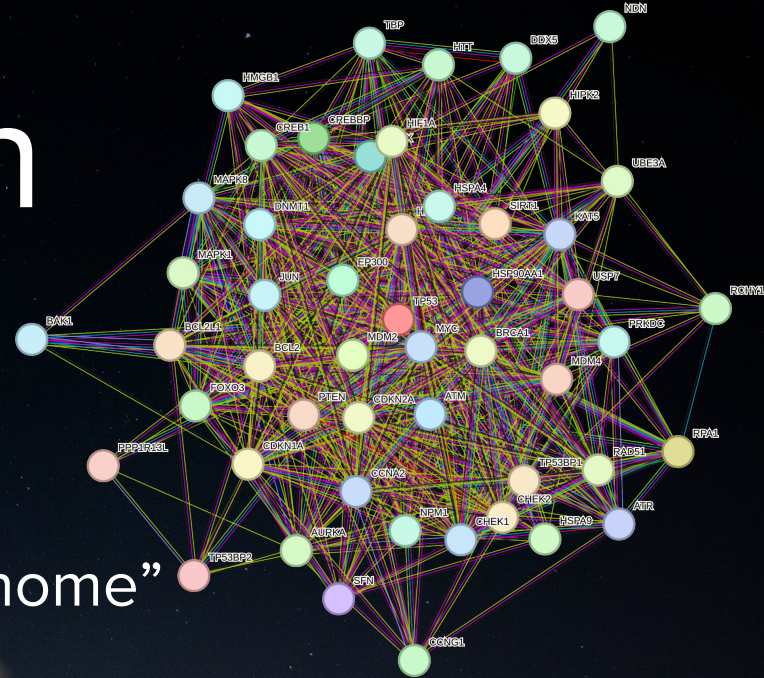


Protein-Protein Interaction:

Analysis of the “Guardian of the Genome”



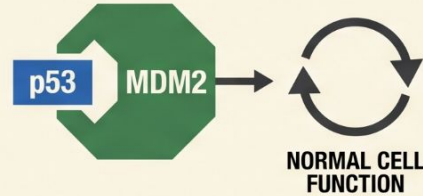
hendra.bunyamin@gmail.com

THE GUARDIAN OF THE GENOME

How TP53 prevents cancer and why it fails.

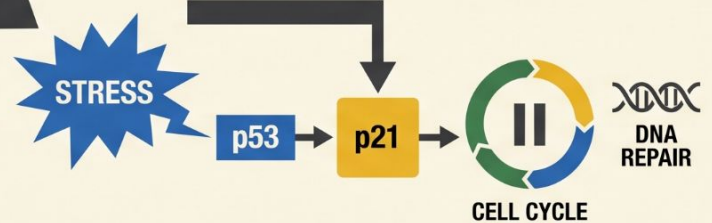
THE GUARDIAN'S DORMANT STATE

Under healthy conditions, inhibitor protein MDM2 keeps p53 levels low for normal cell function.



STEP 1: HALT AND REPAIR

Upon sensing stress, p53 activates p21 to pause the cell cycle for DNA repair.



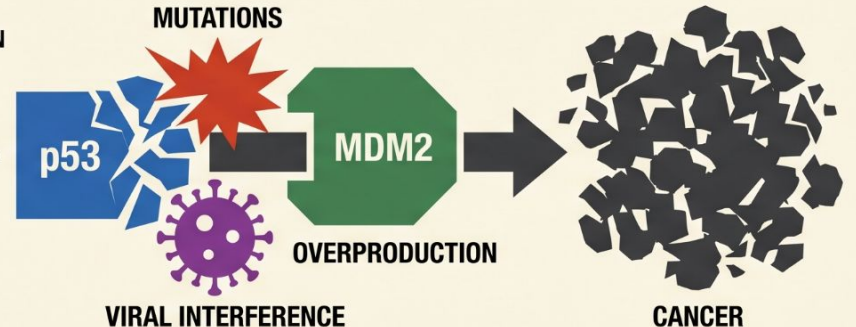
STEP 2: PROGRAMMED SELF-DESTRUCTION

If damage is irreparable, p53 triggers pro-apoptotic proteins like BAX to safely destroy the cell.



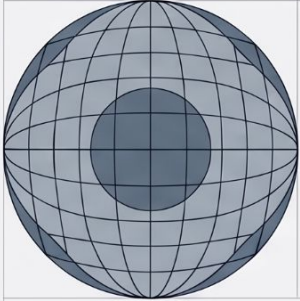
PATHWAY FAILURE AND CANCER

In over 50% of cancers, p53 is inactivated by mutations, MDM2 overproduction, or viral interference.



THE p53 CASCADE: FROM MUTATION TO MALIGNANCY

PHASE I: THE REGULATORY BREAKDOWN



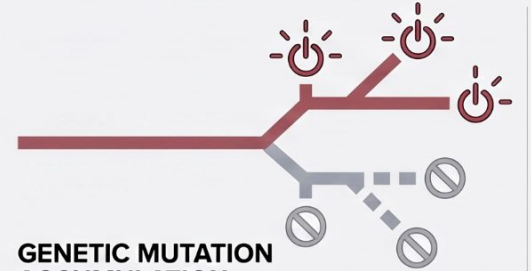
p53 INACTIVATION

Normal cells with DNA damage fail to trigger cell cycle arrest or repair.



S-PHASE REPLICATION

Damaged DNA is erroneously replicated during the S-phase instead of being corrected.



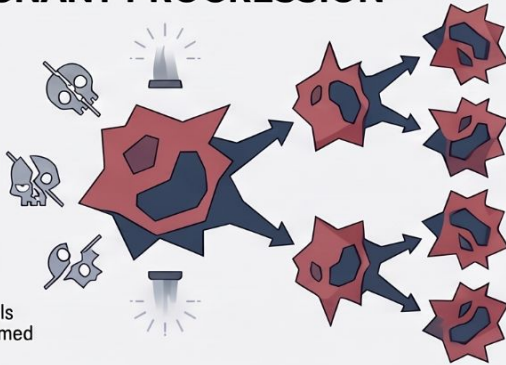
GENETIC MUTATION ACCUMULATION

Oncogenes (KRAS, MYC) activate while tumor suppressors (RB1, BRCA1) are deactivated.

PHASE II: MALIGNANT PROGRESSION

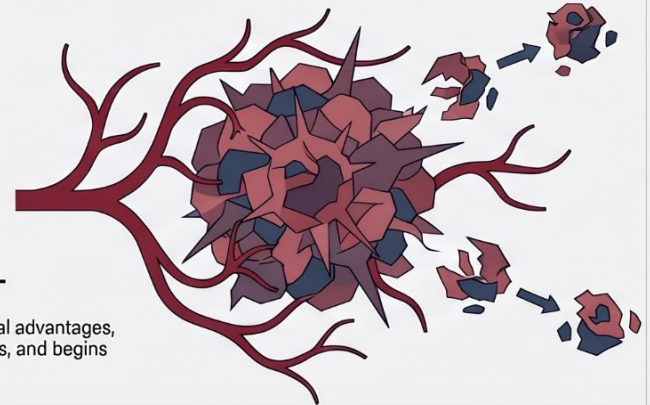
UNCONTROLLED MITOSIS

Mutated cells ignore death signals and successfully evade programmed cell death (apoptosis).



CANCER CELL FORMATION

The cell gains survival advantages, triggers angiogenesis, and begins to metastasize.

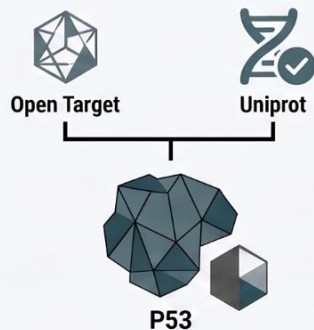


THE BIOINFORMATICS PIPELINE: FROM DATABASE SEARCH TO 3D IDENTIFICATION

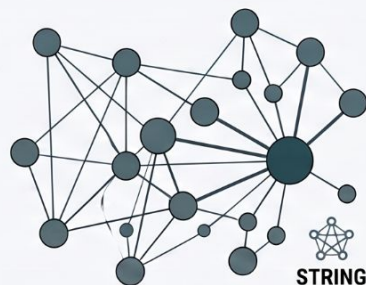
A four-stage computational workflow to identify, map, and analyze key protein targets in cancer research, moving from initial validation to complex 3D structural mapping.

PHASE 1: TARGET DISCOVERY & INTERACTION MAPPING

Database Exploration & Validation



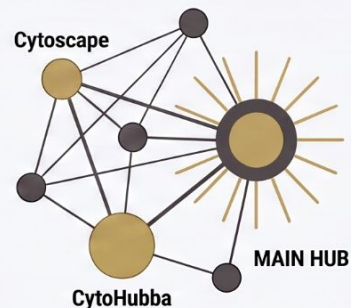
Use Open Target and Uniprot to validate P53 as a key cancer protein.



Construct interaction networks using empirical data and literature via the STRING database.

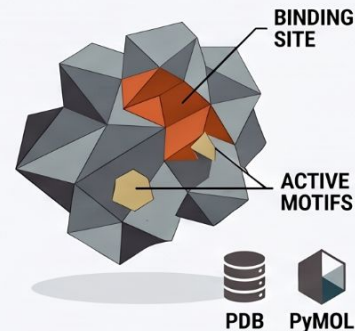
PHASE 2: TOPOLOGICAL ANALYSIS & 3D MODELING

Centrality Computation & Hub Identification



Use Cytoscape and CytoHubba to calculate Degree and Closeness to find main hubs.

3D Structural Identification



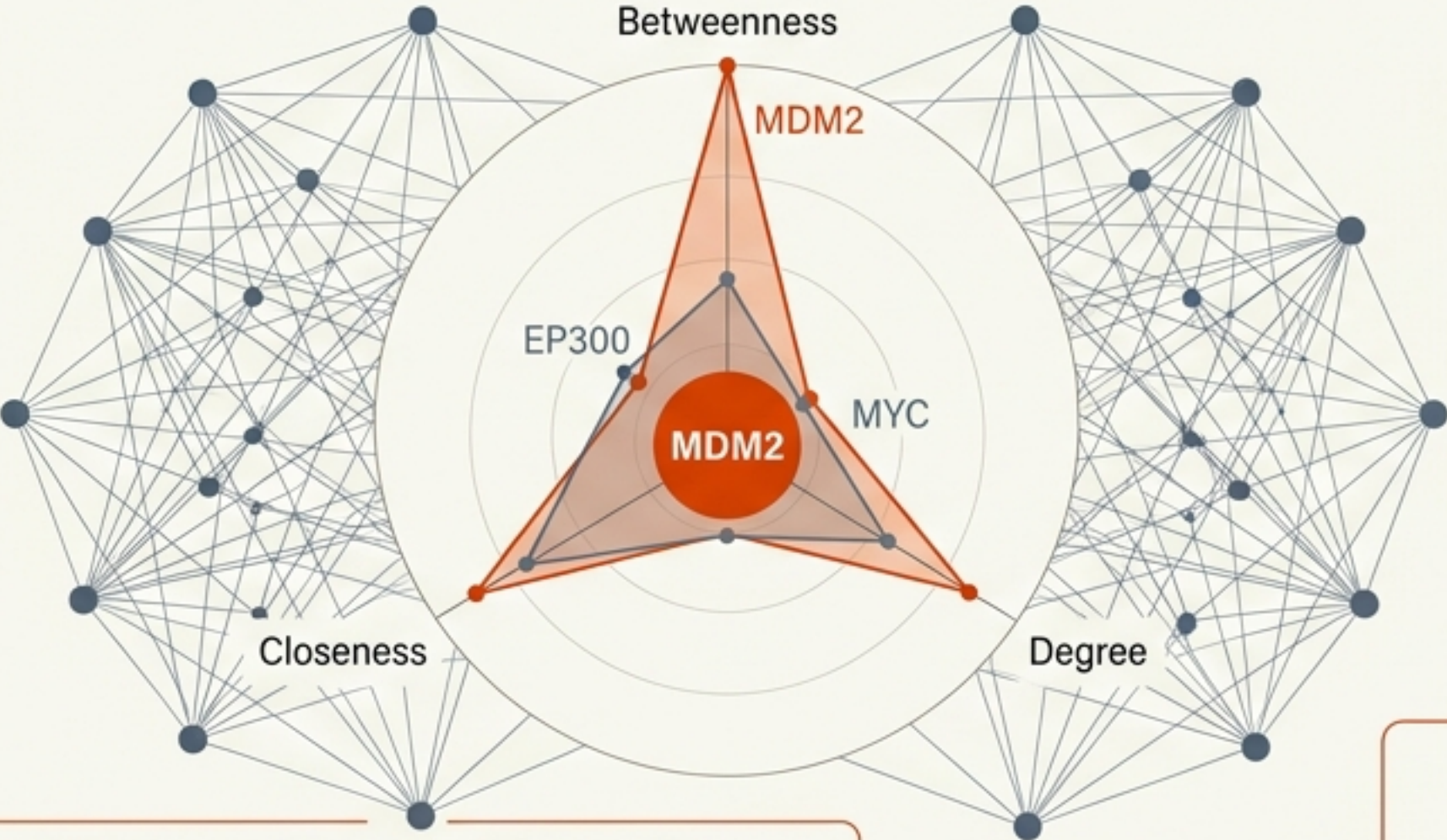
Map binding sites and active motifs using PDB and PyMOL visualization software.

SUMMARY OF TOOLS

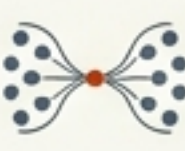
Pipeline Stage	Key Software/Databases
Discovery	Open Target, Uniprot
Mapping	STRING Database
Analysis	Cytoscape, CytoHubba

Topological Analysis Identifies MDM2 as the Critical Network Hub

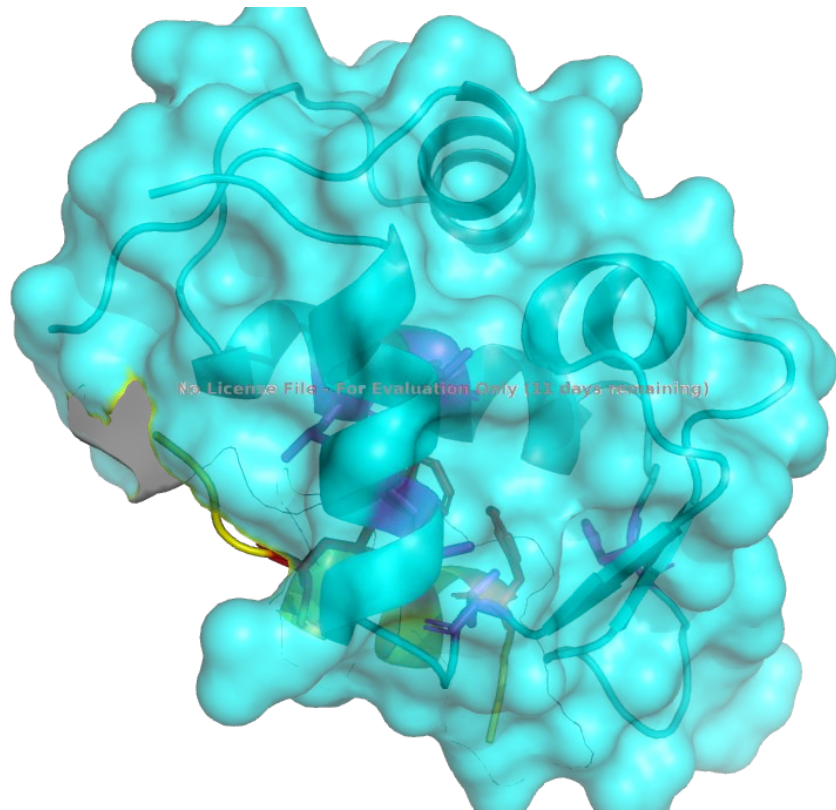
Across Degree, Closeness, and Betweenness metrics, MDM2 emerges as structurally irreplaceable within the target protein network.



By dominating Betweenness, Closeness, and Degree, MDM2 functions as the primary bottleneck and most crucial hub in the network, making it the definitive priority for downstream targeting.

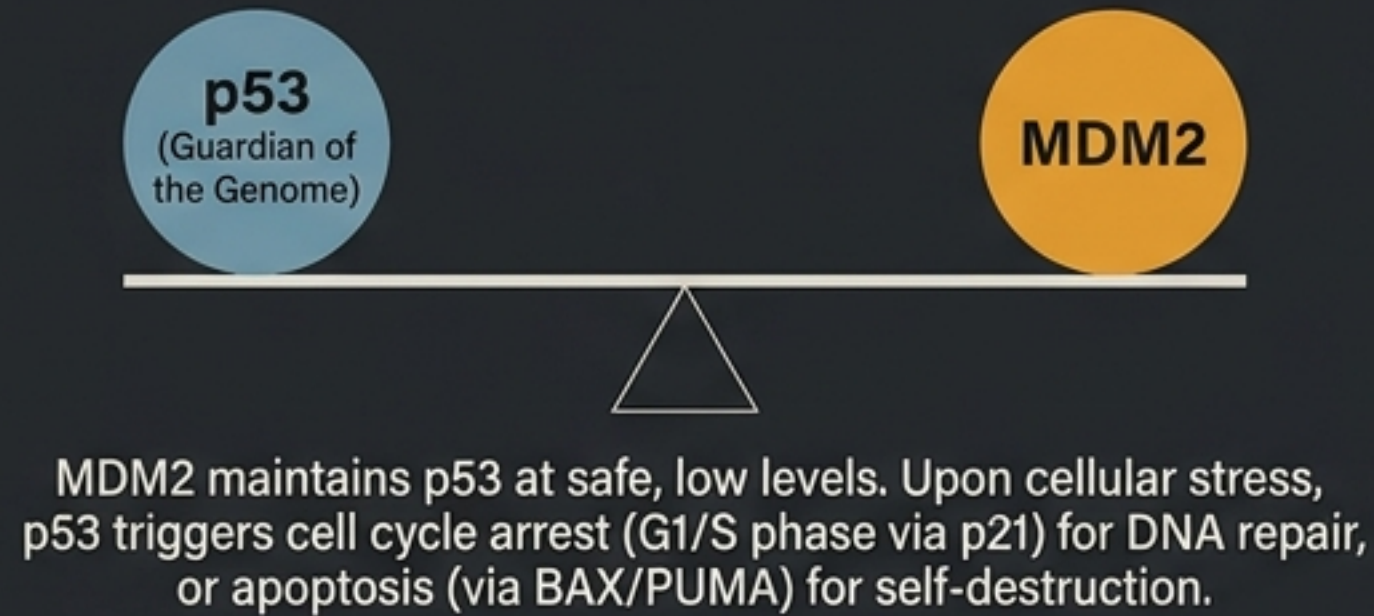
Rank / Node	 Degree	 Closeness	 Betweenness	 Strategic Delta
1: MDM2	43	46.5	77.8467	+56% Betweenness vs. MYC
2: EP300	41	45.5	77.224	--
3: MYC	41	45.5	49.87226	--

3D Binding Site Identification



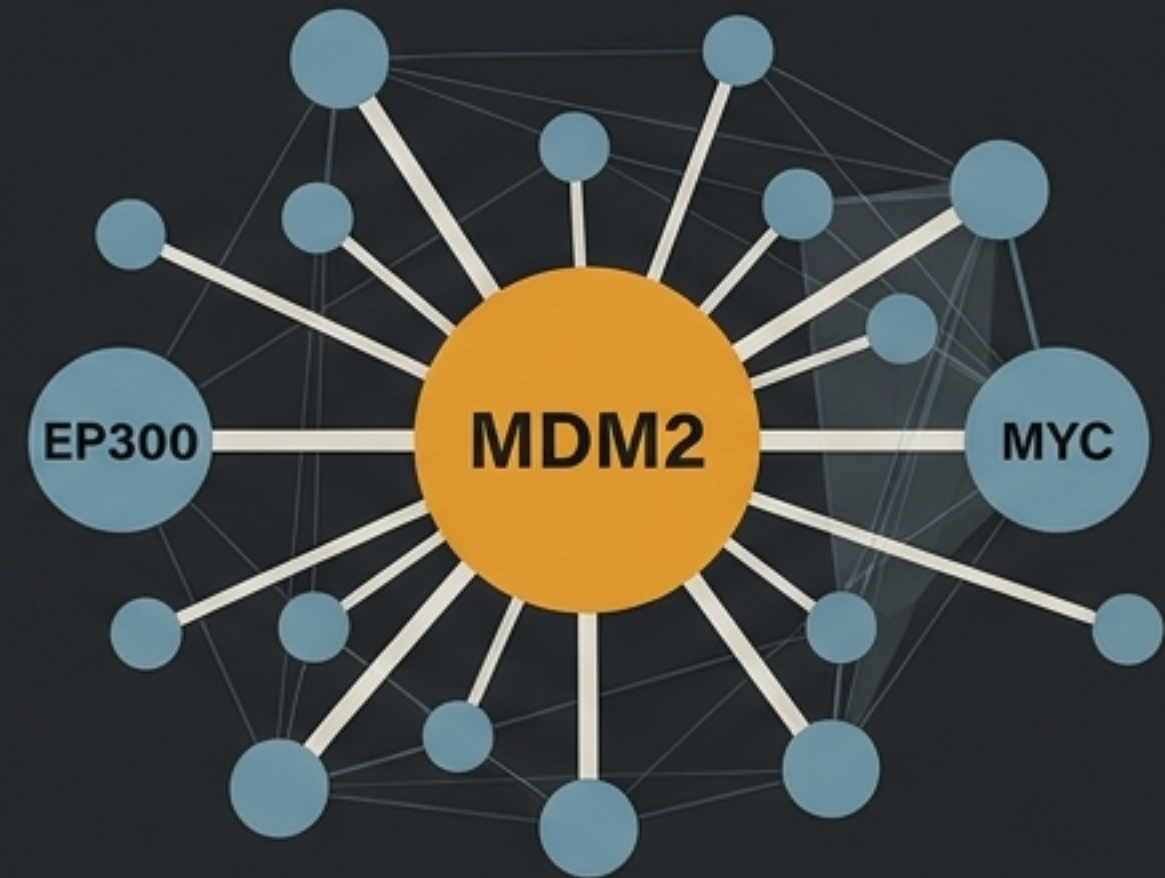
MDM2 acts as both the primary biological **brake** on p53 and the topological epicenter of the cellular network.

The Biological Brake



MDM2 Overexpression tags p53 for premature destruction. Unchecked DNA damage proceeds -> No G1/S arrest -> Uncontrolled mitosis and malignant transformation

The Topological Epicenter



Degree	43	(Rank 1)
Closeness	46.5	(Rank 1)
Betweenness	77.84	(Rank 1 - Dominant)

MDM2 outranks all other critical proteins, including EP300 and MYC, dictating it as the most crucial hub governing the cellular network.

The convergence of biological pathology and network topology exposes MDM2 as the ultimate bottleneck in oncogenesis.

The Pathological Executioner

The TP53 pathway is disrupted in >50% of all human cancers. By overexpressing MDM2, cancer cells systematically dismantle the genome's primary defense system, entirely neutralizing the cell's ability to trigger p21-mediated cell cycle arrest or BAX/PUMA-mediated apoptosis.

The Master Information Hub

CytoHubba topology confirms that MDM2's unrivaled Betweenness (77.84) and Closeness (46.5) scores give it unprecedented structural authority. It controls the flow of information across the protein network more tightly than established oncogenes like MYC or EP300.

**TARGET:
MDM2**

Operating simultaneously as the biological executioner of p53 and the supreme systemic hub, MDM2 dysregulation is the necessary catalyst for the cancer cascade. Consequently, neutralizing overactive MDM2 represents the highest-leverage therapeutic intervention point to restore genomic stability.